Yifei Zhang | Applied Scientist

Amazon Alexa AI, Boston, Massachusetts
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Education

Emory University — **Atlanta, GA**
Ph.D. in Computer Science — *Aug. 2022 – Mar. 2026*
Advisor: Prof. Liang Zhao
Dissertation: Benchmarking, Correcting, and Accelerating AI Reasoning across Diverse Modalities

Columbia University — **New York, NY**
M.S. in Data Science — *Sept. 2020 – Dec. 2021*

Duke University — **Durham, NC**
Master of Engineering Management — *Aug. 2019 – May 2020*

Dalian University of Technology — **Dalian, China**
B.E. in Engineering Mechanics — *Sept. 2015 – June 2019*

Research Focus

AI agents and evaluation; explainable and trustworthy AI; multimodal reasoning.

Develop benchmarks and methods for long-horizon dialogue, memory, visual explanation, retrieval-augmented generation, and reasoning across language, vision, and clinical data. Representative work includes ATOD, Hear2Act, MAGI, ELAD, and Saliency-Bench.

Research and Industry Experience

Amazon — **Boston, MA**
Applied Scientist — *Apr. 2026 – Present*
- Develop scalable evaluation methods for agentic conversational systems, covering reliability, task completion, conversational quality, and multimodal behavior.
- Combine evaluation signals into interpretable assessments for model and system improvement.
- Research privacy-aware model adaptation and distillation under practical system constraints.

Amazon — **Boston, MA**
Applied Scientist Intern — *May 2025 – Aug. 2025*
- Developed a memory system and multi-agent evaluation framework for conversational systems.
- Built synthetic dialogue-generation and benchmarking pipelines for long-horizon, multi-turn, and concurrent user tasks.

Amazon AGI — **Boston, MA**
Research Scientist Intern — *June 2024 – Sept. 2024*
- Developed reward modeling and preference-based evaluation workflows for LLM-driven conversational interactions.
- Built synthetic preference-data generation and parameter-efficient fine-tuning pipelines.

Emory University — **Atlanta, GA**
Ph.D. Student Researcher — *Aug. 2022 – Mar. 2026*
Developed evaluation benchmarks and explanation-guided methods for reliable AI across language, vision, and multimodal data.

Publications

Full publication list: Google Scholar.

Peer-Reviewed and Accepted Papers

Dual Decomposition for Text-to-Image Retrieval and Generation
M. Zhu, S. Cheng, G. Bai, **Yifei Zhang**, and L. Zhao
The 20th Conference of the European Chapter of the Association for Computational Linguistics (EACL), Findings, 2027 (accepted).

Beyond Efficiency: A Systematic Survey of Resource-Efficient Large Language Models
G. Bai, Z. Chai, C. Ling, S. Wang, J. Lu, N. Zhang, T. Shi, Z. Yu, M. Zhu, **Yifei Zhang**, X. Song, C. Yang, Y. Cheng, and L. Zhao
ACM Computing Surveys (CSUR), 2026.

ATOD: An Evaluation Framework and Benchmark for Agentic Task-Oriented Dialogue Systems
Yifei Zhang, H. Nayyeri, R. Khaziev, E. Yilmaz, G. Tur, D. Hakkani-Tür, and H. P. Thadakamalla
The 15th International Joint Conference on Natural Language Processing and the 5th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL-IJCNLP), 2026.

Hear2Act: Benchmarking When Prosody Should Change What an Assistant Does
X. Liu, H. Nayyeri, D. Hakkani-Tür, E. Yilmaz, J.-K. Kim, **Yifei Zhang**, C. Peris, and H. Thadakamalla
Conference on Neural Information Processing Systems (NeurIPS), RTCA Workshop, 2026 (accepted poster).

Saliency-Bench: A Comprehensive Benchmark for Evaluating Visual Explanations
Yifei Zhang, J. Song, S. Gu, T. Jiang, B. Pan, G. Bai, and L. Zhao
The 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Datasets and Benchmarks Track, 2025.

GraphNarrator: Generating Textual Explanations for Graph Neural Networks
B. Pan, Z. Xiong, G. Wu, Z. Zhang, Y. Hu, **Yifei Zhang**, and L. Zhao
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025.

Distilling Large Language Models for Text-Attributed Graph Learning
B. Pan, Z. Zhang, **Yifei Zhang**, Y. Hu, and L. Zhao
The 33rd ACM International Conference on Information and Knowledge Management (CIKM), 2024.

Multi-Prompt Fine-Tuning of Foundation Models for Enhanced Biomedical Image Segmentation
X. Li, **Yifei Zhang**, and L. Zhao
AI for Health Equity and Fairness, Studies in Computational Intelligence, vol. 1164, pp. 83–91, Springer, 2024.

DUE: Dynamic Uncertainty-Aware Explanation Supervision via 3D Imputation
Q. Zhao, **Yifei Zhang**, M. Zhu, S. Gu, Y. Gao, X. Yang, and L. Zhao
The 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024.

ELAD: Explanation-Guided Large Language Models Active Distillation
Yifei Zhang, B. Pan, C. Ling, Y. Hu, and L. Zhao
The 62nd Annual Meeting of the Association for Computational Linguistics (ACL), Findings, 2024.

Visual Attention Prompted Prediction and Learning
Yifei Zhang, B. Pan, S. Gu, G. Bai, M. Qiu, X. Yang, and L. Zhao
The 33rd International Joint Conference on Artificial Intelligence (IJCAI), 2024.

STES: A Spatiotemporal Explanation Supervision Framework
D. Yu, B. Chen, Y. Li, S. Dhakal, **Yifei Zhang**, Z. Liu, M. Zhang, J. Zhang, and L. Zhao
SIAM International Conference on Data Mining (SDM), 2024.

MAGI: Multi-Annotated Explanation-Guided Learning
Yifei Zhang, S. Gu, Y. Gao, B. Pan, X. Yang, and L. Zhao
IEEE/CVF International Conference on Computer Vision (ICCV), 2023.

ESSA: Explanation Iterative Supervision via Saliency-Guided Data Augmentation
S. Gu, **Yifei Zhang**, Y. Gao, X. Yang, and L. Zhao
The 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2023.

Preprints

Agentic Commerce: A Survey of How AI Agents Are Reshaping Commerce
Yifei Zhang, B. Pan, M. Zhu, J. Pei, and L. Zhao
TechRxiv preprint, 2026.

EmoLLM: Appraisal-Grounded Cognitive-Emotional Co-Reasoning in Large Language Models
Yifei Zhang, M. Li, H. Gao, and L. Zhao
arXiv:2603.16553, 2026.

Explainable Cross-Disease Reasoning for Cardiovascular Risk Assessment from Low-Dose Computed Tomography
Yifei Zhang, J. Zhang, M. Safari, X. Yang, and L. Zhao
arXiv:2511.06625, 2025.

MEGL: Multimodal Explanation-Guided Learning
Yifei Zhang, T. Jiang, B. Pan, J. Wang, G. Bai, and L. Zhao
arXiv:2411.13053, 2024.

Saliency-Guided Hidden Associative Replay for Continual Learning
G. Bai, Q. Zhao, X. Jiang, **Yifei Zhang**, and L. Zhao
arXiv:2310.04334, 2023.

SurroCBM: Concept Bottleneck Surrogate Models for Generative Post-hoc Explanation
B. Pan, Z. Liu, **Yifei Zhang**, and L. Zhao
arXiv:2310.07698, 2023.

Controllable Data Generation Via Iterative Data-Property Mutual Mappings
B. Pan, M. Qin, S. Wang, **Yifei Zhang**, and L. Zhao
arXiv:2310.07683, 2023.

Academic Service

Journal Reviewer: ACM TIST; IEEE TBD; IEEE TNNLS; JACMP; Expert Systems with Applications; Information Processing and Management; Neurocomputing; Information Systems Journal.
Conference Reviewer / Program Committee: ICLR 2025–2027; NeurIPS 2026; AAAI 2027; CVPR 2025–2026; ICCV 2025; ECCV 2026; BMVC 2026; ACL ARR 2025–2026; KDD Research Track 2025–2026; KDD Datasets and Benchmarks Track 2025–2026; MemAgents Workshop (ICLR 2026).
Workshop Organization: Publicity Chair, DeepSpatial 2024 (KDD); Student Co-organizer, FM4sBio 2025 (AAAI).

Honors, Grants, and Memberships

- Sigma Xi, Full Membership.
- Tinker Research Grant.
- KDD 2023 Student Travel Award.
- Dalian University of Technology Overseas Alumni Association Seagle Scholarship Award.
- Academic Excellence Scholarship Award, Dalian University of Technology.
- Technological Innovation Scholarship Award, Dalian University of Technology.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, Shell
AI / LLM Frameworks: PyTorch, Transformers, TensorFlow, vLLM, verl, TRL, PEFT
Training and Infrastructure: Accelerate, DeepSpeed, AWS, Docker, Linux, Git